

ACA 2024 Program

Tuesday 25, 11:30-12:30	Special Session: <i>Computer Algebra for Digital Image Processing and Computer Vision.</i> Session Chair: <i>Marta Lourdes Baguer.</i>
<u>Ángela M. León Mecías</u>	Numerical and Image Analysis research group: lines of research, some results and work in progress.
Eric Martín-García and <u>Damian Valdés Santiago</u>	One-dimensional pattern detection using optimized and matched wavelets.
Tuesday 25, 14:00-15:30	Special Session: <i>Computer Algebra in Education.</i> Session Chair: <i>Michael Wester</i>
<u>Valentina Badía Albanés</u>	The study of uniform convergence in the representation of functions using Computer Algebra Systems.
<u>Włodzimierz Wojas</u>	Can Chat GPT be a useful didactic tool in teaching and learning advanced mathematical analysis optimization topics?
Noah Dana-Picard and Zoltán Kovács	Estrella Solitaria in offset curves.
Tuesday 25, 14:00-15:30	Special Session: <i>Computer Algebra for Digital Image Processing and Computer Vision.</i> Session Chair: <i>Alejandro Mesejo Chiong.</i>
Daniel García Pérez, <u>Marta Baguer Díaz-Romañach</u> and Manuel Vilas Valiente	Image inpainting using the smooth ordering of its patches: two new applications.
Nigreisy Montalvo Zulueta, <u>Ángela M. León Mecías</u> and Lorena León Arencibia	Texture analysis for breast mass malignancy classification in mammography images.
Lázaro Jesús Suárez Nuñez, Alba Álvarez González and <u>Damian Valdés Santiago</u>	Computational morphological analysis of digital images from irradiated tobacco seedlings.
Tuesday 25, 16:00-16:30	Special Session: <i>Computer Algebra for Digital Image Processing and Computer Vision.</i> Session Chair: <i>Damian Valdés Santiago</i>
<u>José Alejandro Mesejo-Chiong</u> and Leonardo Ulloa-Ferrer	3D reconstruction of Diabetic Foot Ulcers to evaluate healing progress.
Wednesday 26, 9:00-9:30	Special Session: <i>Computer Algebra in Coding Theory and Cryptography.</i> Session Chair: <i>Mijail Borges</i>
<u>Daniel Cabarcas</u> , Giulia Gaggero, Elisa Gorla	The complexity of the SupportMinors Modeling for the MinRank Problem.
Wednesday 26, 9:30-10:30	Plenary Talk Chair: <i>Osmani Tito Corrioso</i>
Mijail Borges Quintana	Study of algebraic and computational aspects of the Gröbner representations of linear codes.
Wednesday 26, 11:00-12:30	Special Session: <i>Computer Algebra in Coding Theory and Cryptography.</i> Session Chair: <i>Osmani Tito Corrioso/Mijail Borges</i>
Osmani Tito-Corrioso, Mijail Borges-Quintana, Miguel A. Borges-Trenard, Omar Rojas, Guillermo Sosa-Gómez	Combined General Methodologies of Key Space Partition for the Cryptanalysis of Block Ciphers.
<u>Ernesto Dominguez Fiallo</u> and Pablo Freyre Arrozarena	On the security of digital signature schemes based on CFS-like codes.
<u>Mijail Borges-Quintana</u> , I. Alvarez-Barrientos, Miguel A. Borges-Trenard, E. Martínez Moro and J. A. Ornella Rodríguez	Complete Gröbner Basis for Linear and Lattice Codes.

Thursday 27, 9:00-10:30	Special Session: <i>Computer Algebra in Education.</i> Session Chair: <i>Michael Wester</i>
S. Bahrami and <u>David J. Jeffrey</u>	Ecstatic about extactic.
<u>Michel Beaudin</u>	Choosing an antiderivative is not easy!
J. Middeke, A. Olagunju and <u>David J. Jeffrey</u>	Database for Orthogonal matrices.
Thursday 27, 11:00-12:30	Special Session: <i>Integration and ODEs: Implementation and Applications.</i> Session Chair: <i>Michael Wester</i>
<u>David J. Jeffrey</u> and David Stoutemyer	Albert Rich 1949-2023
<u>Michel Beaudin</u>	Do like Rubi : don't be afraid of using arctanh !
A. C. Norman and <u>David J. Jeffrey</u>	A Future for Rubi?
Thursday 27, 11:00-12:30	Special Session: <i>Machine Intelligence and Symbolic Computation.</i> Session Chair: <i>Bo Huang</i>
<u>Yang-Hui He</u>	The AI Mathematician: From Physics, to Geometry, to Number Theory.
<u>Rui-Juan Jing</u>	A Dataset for Suggesting Variable Orderings for Cylindrical Algebraic Decompositions.
<u>Dongchen Jiang</u>	Formal Representation of Mathematical Texts.
Thursday 27, 14:00-14:30	Special Session: <i>Integration and ODEs: Implementation and Applications.</i> Session Chair: <i>Michel Beaudin</i>
<u>Patrick Scheibe</u> via Michel Beaudin- VIDEO	Rubi: Solving Integrals, One Rule at a Time.
Thursday 27, 14:00-14:30	Special Session: <i>Machine Intelligence and Symbolic Computation.</i> Session Chair: <i>Jing Yang</i>
<u>Weifeng Shang</u>	A Framework of Plain Geometric Knowledge Discovery.
Thursday 27, 14:30-15:30	ACA 2024 - Panel Discussion Chair: <i>Jing Yang</i>
Friday 28, 9:00-10:30	General Session Session Chair: <i>Michael Wester</i>
<u>Dingkang Wang</u> and Xiaopeng Zheng	Completing Parametric Unimodular Rows to Unimodular Matrices.
<u>Bo Huang</u>	Algebraic Analysis of Zero-Hopf Bifurcations of Polynomial Differential Systems.
Hoon Hong and <u>Jing Yang</u>	Computing the greatest common divisor of several parametric univariate polynomials via generalized subresultant polynomials.